

Online Medical Test Result System

A step-by-step guide to using the CRUTECH (UNICROSS) Medical Center test result platform — for administrators and laboratory technologists.

[Next.js](#) · [MongoDB](#) · [NextAuth](#)

[Admin & Technologist Roles](#)

[Secure Result Links](#)

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Overview

What the system does and who it is for.

The Online Medical Test Result System replaces paper-based handling of laboratory test results at CRUTECH (UNICROSS) Medical Center. It lets laboratory staff capture patient records, conduct and process tests online, and deliver results securely — without requiring patients to create an account.

The system implements four core objectives:

Objective	How the system delivers it
Capture & store patient records	Every test is linked to a patient record with a unique Patient ID, stored in MongoDB.
Online search & retrieval	Staff can search patients and test records instantly by name, ID, or test type.
Secure online access to results	A tokenized link is generated when a result is sent — no separate patient account is needed.
Confidentiality & data protection	Role-based access (Admin/Technologist), hashed passwords, and an identity check (last name + date of birth) before any result is revealed.

User Roles & Use Cases

Two staff roles, one public (no-login) result flow.

ADMIN

Administrator

Manages staff accounts and oversees every test record in the system.

- Add & modify staff accounts (Admin or Technologist)
- View every test record across all technologists
- Share a result's secure link
- Print a clean, letterhead result report

TECHNOLOGIST

Technologist

Runs the day-to-day testing workflow, patient by patient.

- Modify their own account details
- Conduct a test (register patient + test type)
- Process a test (enter result parameters)
- View their own test history
- Send a completed result (generates the secure link)

There is a third, unauthenticated use case: a **patient (or anyone with the link)** can open a sent result at `/results/view/[token]`, confirm their last name and date of birth, and view/print the report — with no account required.

Demo Login Credentials

Use these accounts to explore the live application yourself.

Admin	EMAIL	PASSWORD
	demo.admin@example.com	Demo@12345
Technologist	EMAIL	PASSWORD
	demo.tech@example.com	Demo@12345

Live application: <https://medical-test-result-system.vercel.app>

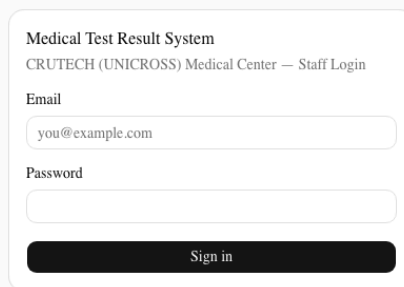
Note: these are separate demo accounts seeded specifically for this guide — they are not the production administrator account. Demo data (a sample patient, a sample test type, and one sent result) is pre-loaded so every screen in this guide has something to show. Feel free to add, edit, or delete demo records while exploring.

Administrator Guide

Sign in, manage staff, and oversee every test in the system.

1 Sign In

Go to the login page and enter the admin email and password from Section 3. The system detects your role automatically and takes you to the Admin Dashboard.

A mockup of a staff login page. It features a white login card centered on a light gray background. The card has a title 'Medical Test Result System' and a subtitle 'CRUTECH (UNICROSS) Medical Center — Staff Login'. Below these are labels for 'Email' and 'Password'. The email input field contains the text 'you@example.com'. The password input field is empty. At the bottom of the card is a dark gray button with the text 'Sign in' in white.

Medical Test Result System
CRUTECH (UNICROSS) Medical Center — Staff Login

Email

you@example.com

Password

Sign in

Fig. 4.1 — Staff login page

2 Admin Dashboard

The dashboard gives an at-a-glance count of staff accounts, patients, and tests by status (in progress, awaiting send, sent) — a quick health check of activity.

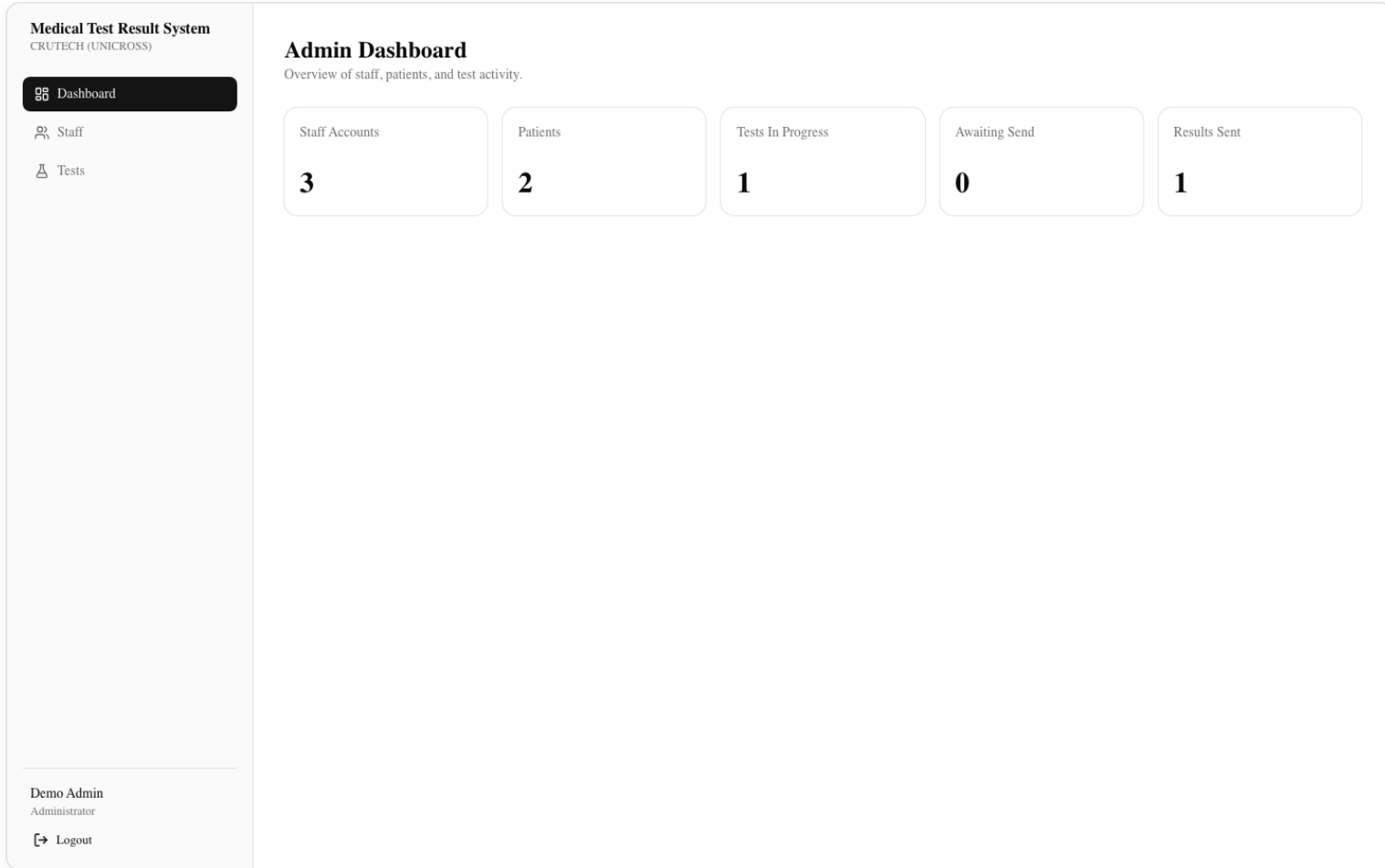


Fig. 4.2 — Admin Dashboard overview

Managing Staff Accounts

1 Open Staff

Select **Staff** in the sidebar to see every admin and technologist account, their role, and whether they're active.

The screenshot shows the 'Medical Test Result System' interface. On the left is a sidebar with the system name 'CRUTECH (UNICROSS)' and three menu items: 'Dashboard' (with a grid icon), 'Staff' (with a group of people icon and highlighted in dark blue), and 'Tests' (with a flask icon). At the bottom of the sidebar, it shows the user 'Demo Admin Administrator' and a 'Logout' button. The main content area is titled 'Staff' with a subtitle 'Manage admin and technologist accounts.' and an '+ Add Staff' button. Below this is a table with five columns: Name, Email, Role, Status, and Actions. The table contains three rows of staff members.

Name	Email	Role	Status	Actions
Demo Technologist	demo.tech@example.com	Technologist	Active	Modify
Demo Admin	demo.admin@example.com	Admin	Active	Modify
System Administrator	admin@example.com	Admin	Active	Modify

Fig. 4.3 — Staff list

2 Add a Staff Account

Click **Add Staff**, fill in the name, email, an optional phone number, choose the role (Admin or Technologist), and set a temporary password. Share that password with the new staff member so they can log in and change it under **My Account**.

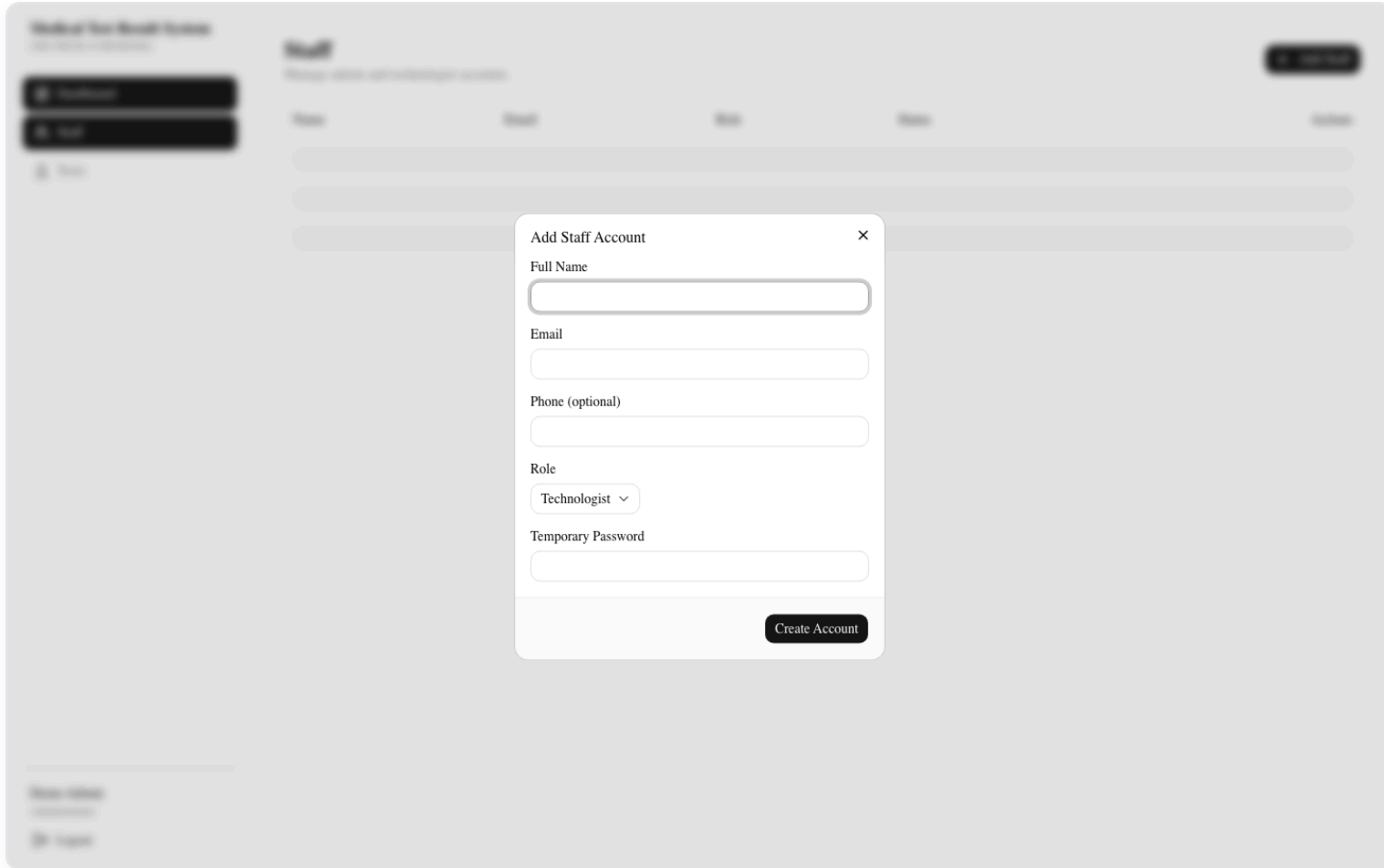


Fig. 4.4 — Add Staff Account dialog

Use the **Modify** button on any row to update a staff member's name, phone, active status, or reset their password.

Viewing, Sharing & Printing Tests

1 Browse All Tests

The **Tests** page lists every test record system-wide, searchable by patient name, patient ID, or test type — regardless of which technologist conducted it.

Medical Test Result System
CRUTECH (UNICROSS)

Dashboard

Staff

Tests

Demo Admin
Administrator

Logout

Tests

View, share, and print all test records across staff.

Search by patient, patient ID, or test type...

Patient	Test Type	Conducted By	Status	Date	Actions
Grace Bassey (PT-00002)	Full Blood Count	Demo Technologist	Pending	9/21/2026	Share Print
John Adewale (PT-00001)	Full Blood Count	Demo Technologist	Sent	9/21/2026	Share Print

Fig. 4.5 — All test records

2 Share or Print a Result

The share icon copies a sent result's secure link to the clipboard, ready to forward to the patient by any channel. The print icon opens a clean, letterhead-style report — click **Print** to save as PDF or send to a printer.

Medical Test Result System
CRUTECH (UNICROSS)

Dashboard

Staff

Tests

Demo Admin
Administrator

Logout

Print

CRUTECH (UNICROSS) MEDICAL CENTER

Medical Test Result Report

Patient Name

John Adewale

Patient ID

PT-00001

Date of Birth

3/12/1998

Gender

Male

Test Type

Full Blood Count

Conducted By

Demo Technologist

Date

9/21/2026

Status

Sent

Results

Parameter	Value	Unit	Normal Range	Flag
Hemoglobin	14.2	g/dL	13-17	Normal
White Blood Cell Count	11.8	x10^9/L	4-11	High
Platelet Count	250	x10^9/L	150-400	Normal

Notes

Routine annual checkup

Fig. 4.6 — Printable result report

Technologist Guide

Conduct → Process → Send — the day-to-day workflow.

1 Dashboard & My Tests

After signing in with a technologist account, the dashboard summarizes your own test activity by status. **My Tests** lists every test you've conducted.

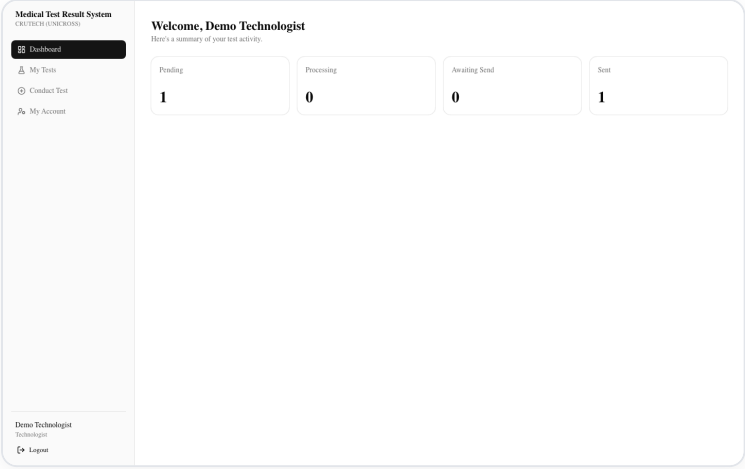


Fig. 5.1 — Technologist Dashboard

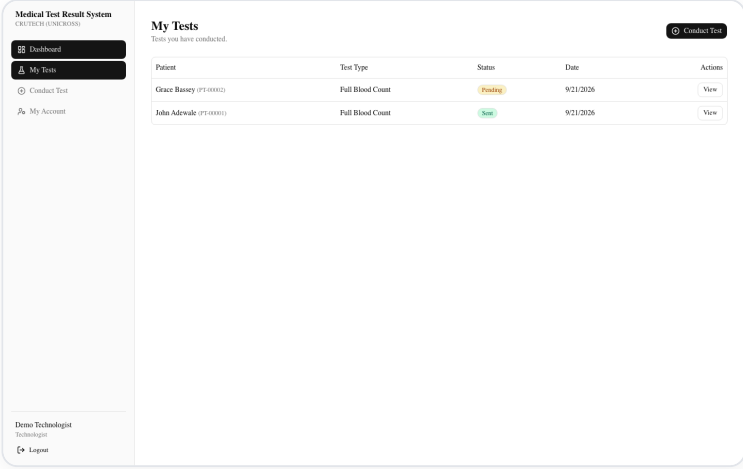


Fig. 5.2 — My Tests list

Conducting a Test

1 Select or Add a Patient, Choose a Test Type

Click **Conduct Test**. Search for an existing patient by name or ID, or use **New Patient** to register one on the spot. Pick a test type from the list, or add a new one with **New Test Type**. Add optional notes, then **Start Test Record** — this creates the record with status *Pending*.

Medical Test Result System
CRUTECH (UNICROSS)

Dashboard

My Tests

Conduct Test

My Account

Demo Technologist
Technologist
Logout

Conduct Test

Select a patient and test type to open a new test record.

Patient

New Patient

Grace

Grace Bassey (PT-00002)

Test Details

New Test Type

Test Type

Full Blood Count

Notes (optional)

Start Test Record

Fig. 5.3 — Conduct Test: patient + test type selection

Processing & Sending Results

1 Enter Result Parameters

Open a test from **My Tests**. Add one row per parameter (e.g. Hemoglobin, White Blood Cell Count) with its value, unit, normal range, and a flag (Normal / Low / High / Critical). Add notes if needed, then **Save Results** — status moves to *Completed*.

Medical Test Result System
CRUTECH (UNICROSS)

Dashboard

My Tests

Conduct Test

My Account

Demo Technologist
Technologist

Logout

Grace Bassey (PT-00002)

Pending

Full Blood Count

Patient Details

Date of Birth
11/2/2001

Gender
Female

Phone
08087654321

Email
grace.bassey@example.test

Results

+ Add Parameter

Parameter	Value	Unit	Normal Range	Flag	
e.g. Hemoglobin				Normal	

Notes

Follow-up test

Save Results

Fig. 5.4 — Entering result parameters

2 Send the Result

Once completed, click **Send Result**. This generates a unique secure link and moves the record to *Sent* — the link is shown on-screen (and copyable) for you to forward to the patient.

Medical Test Result System
CRUTECH (UNICROSS)

Dashboard

My Tests

Conduct Test

My Account

Demo Technologist
Technologist

Logout

John Adewale (PT-00001)

Sent

Full Blood Count

Patient Details

Date of Birth
3/12/1998

Gender
Male

Phone
08012345678

Email
john.adewale@example.test

Results

Parameter	Value	Unit	Normal Range	Flag
Hemoglobin	14.2	g/dL	13-17	Normal
White Blood Cell Count	11.8	x10 ⁹ /L	4-11	High
Platelet Count	250	x10 ⁹ /L	150-400	Normal

Notes

Routine annual checkup

Result Link

https://medical-test-result-system.vercel.app/results/view/demo1234567890abcdef1234567890abcdef12345678

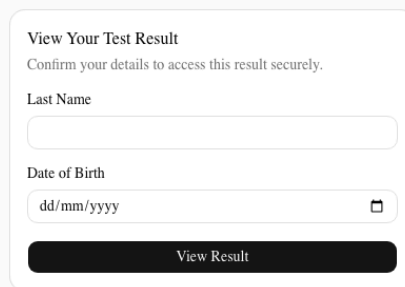
Fig. 5.5 — Completed record with generated result link

Patient Result Access (Public Link)

No account, no password — just the link and a quick identity check.

1 Open the Link & Verify Identity

Anyone who opens a sent result's link is asked for the patient's last name and date of birth. This confirms the right person is viewing the result, without requiring account creation or a password.

A screenshot of a web form titled "View Your Test Result". Below the title is a subtitle "Confirm your details to access this result securely." The form contains two input fields: "Last Name" with a text input box, and "Date of Birth" with a date picker showing "dd/mm/yyyy" and a calendar icon. At the bottom is a dark button labeled "View Result".

View Your Test Result
Confirm your details to access this result securely.

Last Name

Date of Birth
 

View Result

Fig. 6.1 — Identity verification gate

2 View & Print the Result

Once verified, the full report is displayed — patient details, test type, all result parameters with flags, and any notes — with a **Print** button to save or print a copy. A wrong name or date of birth is rejected with no details revealed.

CRUTECH (UNICROSS) MEDICAL CENTER

Medical Test Result Report

Patient Name John Adewale	Patient ID PT-00001
Date of Birth 3/12/1998	Gender Male
Test Type Full Blood Count	Conducted By Demo Technologist
Date 9/21/2026	Status Sent

Results

Parameter	Value	Unit	Normal Range	Flag
Hemoglobin	14.2	g/dL	13-17	Normal
White Blood Cell Count	11.8	$\times 10^9/L$	4-11	High
Platelet Count	250	$\times 10^9/L$	150-400	Normal

Notes

Routine annual checkup

Fig. 6.2 — Verified, printable test result

Quick Reference & Troubleshooting

Situation	What to do
Forgot a staff password	An Admin can reset it from Staff → Modify and set a new temporary password.
Technologist wants to change their own password	Go to My Account and enter a new password.
Need a new test type (e.g. "Malaria Parasite Test")	Click New Test Type from the Conduct Test screen.
Patient lost their result link	An Admin can re-copy the link from the Tests page's share icon — the link does not expire or change.
Wrong last name / DOB on the public page	Access is denied with no data shown. Double-check spelling and date format (YYYY-MM-DD).
A test won't send	Results must be saved (status <i>Completed</i>) before Send Result becomes available.

Status	Meaning
<i>Pending</i>	Test record created; results not yet entered.
<i>Completed</i>	Results saved; ready to send.
<i>Sent</i>	Secure link generated; patient can view the result.